

7 *Use of the Data Booklet is relevant to this question.*

Sodium percarbonate, $(\text{Na}_2\text{CO}_3)_x \cdot y(\text{H}_2\text{O}_2)$, is an oxidising agent used in laundry cleaning products.

On acidification, 10.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ sodium percarbonate releases 48.0 cm^3 of carbon dioxide at room temperature and pressure.

An identical sample, on titration with $0.0500 \text{ mol dm}^{-3} \text{ KMnO}_4$, requires 24.0 cm^3 before the first pink colour appears. 2 moles of KMnO_4 reacts with 5 moles of H_2O_2 .

What is the ratio of y/x ?

A $\frac{1}{3}$

B $\frac{2}{3}$

C $\frac{3}{2}$

D 3